VENKAT AKHIL ANKEM

Operations Research · Large-Scale Optimization · Applied Machine Learning for Industrial Systems

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PROFESSIONAL SUMMARY

Ph.D. candidate in Industrial Engineering (Operations Research) at Polytechnique Montréal / GERAD, with a three-year MITACS-funded industrial collaboration inside Rio Tinto's mine-planning operations (2022–2025) and continued monthly advisory ties with the team since. Delivered cutting-plane and rolling-horizon methods achieving a 90% reduction in mine-scheduling computation time and an 85% reduction in time-to-feasibility, applied on real-life mine instances. Published in Computers & Operations Research (2026); second paper (major revision) at Engineering Optimization; 5 peer-reviewed journal publications in total. Current doctoral work extends this research by integrating machine learning (GNN) into MILP solution strategies. Developing instance-difficulty prediction and solver-guidance models on large-scale benchmark datasets built from Rio Tinto's mine models (100,000+ variable instances). Combines rigorous mathematical optimization with applied ML, deep reinforcement learning and data science to deliver measurable, production-grade operational impact; seeking to bring this combination to an industrial Data & Artificial Intelligence team on a full-time basis following Ph.D. completion (targeted defense April–May 2027).

CORE COMPETENCIES

Mathematical Optimization	Algorithms & Heuristics	Applied ML, Data & Software
Mixed-Integer Programming (MILP/MIP)	Large Neighborhood Search (LNS)	PyTorch, Graph Neural Networks
Cutting Planes & Model Strengthening	Relax-and-Repair Frameworks	Deep RL (PPO, DQN, Q-learning), scikit-learn
LP Relaxations, Branch-and-Cut	Rolling-Horizon Decomposition	C++, Python, R, SQL, SAS
Combinatorial Optimization	ML-Guided Optimization	Gurobi, CPLEX, Xpress, Pyomo, Git

PROFESSIONAL EXPERIENCE

Graduate Researcher — Operations Research (MITACS Industrial Collaboration) — Rio Tinto, Montréal, QC *Jul. 2022 – Sep. 2025*

- Achieved a 90% reduction in mine-scheduling computation time via novel cutting planes and rolling-horizon heuristics deployed on Rio Tinto's industrial planning instances.
- Reduced time-to-feasibility by 85% through constraint relaxation strategies, and novel heuristic framework enabling faster initial solution generation for large-scale MIP models used in production planning.
- Worked directly with the industrial partner to translate research prototypes into production-relevant algorithmic tools, bridging academic OR with live mining operations and existing data infrastructure.
- Formal MITACS engagement concluded Sep. 2025; continue monthly technical check-ins with the Rio Tinto team through present, advising on scheduling model deployment and next-stage research direction.

Doctoral Researcher — Operations Research — Polytechnique Montréal · GERAD, Montréal, QC *Aug. 2021 – Present*

- Designed novel cutting planes for the Open-Pit Mine Production Scheduling Problem (OPMPSP), tightening LP relaxation bounds and reducing branch-and-bound search on 100,000+ variable instances.
- Developed relax-and-repair and parallel warm-start hybrid heuristic frameworks in C++ (OpenMP) for scalable feasible-solution generation and improvement (second paper, major revision, Engineering Optimization).
- Third project (in progress): Integrate GNN models with optimization process to guide the solver strategy in MILP, trained on large-scale benchmark datasets generated via structured augmentation strategies across the real-life mine instances.
- Presented research at JOPT 2025/2026 (Montréal), ISMP 2024 (Montréal), and IISE Annual Conference 2024.
- Author: "New cutting planes for open-pit mine scheduling." Computers & Operations Research, 2026.

Trainee Decision Scientist — Mu Sigma Inc., Bengaluru, India *Sep. 2015 – Jul. 2016*

- Delivered end-to-end analytics solutions for U.S. pharmaceutical and retail clients in a fast-paced, client-facing agile environment.
- Built a physician targeting tool in SAS, SQL and R to identify and prioritize over 500 high-potential doctors, enabling sales representatives to focus outreach and drive brand conversion from competitor medications.
- Built and maintained analytics pipelines and database management systems supporting recurring reporting and ad-hoc client requests across multiple accounts.
- Partnered directly with client stakeholders and internal cross-functional teams to translate open-ended business questions into structured analytical solutions and decision-support tools.

Research Intern — Flight Safety Analytics — École Nationale de l'Aviation Civile (ENAC), Toulouse, France *Jun. 2020 – Jan. 2021*

- Developed a Python flight-safety analysis tool; applied data analysis and ML (random forest, regression) to Be58/TB20 and A340-300 flight datasets to identify unsafe operational conditions.

Graduate Research Assistant — Aerospace Materials — Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea *Sep. 2016 – Jul. 2018*

- Experimental and simulation-based research on aerospace composite structures (hypervelocity impact, morphing wing materials). Co-authored 3 peer-reviewed journals; 4 international conference presentations.

SELECTED PROJECTS (GITHUB.COM/VENKAT-AKHIL-ANKEM)

- Mine Production AI Optimizer — GNN/ML pipeline for MILP difficulty prediction and solver guidance on open-pit scheduling instances.
- ML Scheduling Difficulty Predictor — supervised learning models (scikit-learn) predicting solve time and difficulty from instance structure.
- PPO Mine Block Sequencing — deep RL agent built from scratch in PyTorch (PPO actor-critic, GAE, action masking over precedence-feasible blocks, curriculum learning) that learns NPV-maximising extraction sequences under MineLib-format precedence and per-period capacity constraints; benchmarked against greedy baselines and an LP-relaxation bound.
- DQN CartPole — Deep Q-Network (Mnih et al., 2015) implemented from scratch in PyTorch/Gymnasium with experience replay, target network, ε-greedy exploration, Huber loss and gradient clipping; unit-tested with pytest.
- RL Job Scheduler — tabular Q-learning agent for parallel machine scheduling.
- Research Paper AI Assistant — LLM-based tool supporting literature review and academic writing workflows.

EDUCATION

Ph.D. in Industrial Engineering (Operations Research) — Polytechnique Montréal · GERAD *Aug. 2021 – Present (expected 2027)*
Supervisors: Prof. Guy Desaulniers & Prof. Michel Gamache · Dissertation: Optimization methods for open-pit mine production planning · MITACS Fellow · Industrial partner: Rio Tinto

M.Sc. in International Air Transport Operations Management — École Nationale de l'Aviation Civile (ENAC), Toulouse, France *Aug. 2018 – Aug. 2021*
OR in aviation: crew scheduling, route assignment, maintenance scheduling · GIFAS Scholar

M.Sc. in Aerospace Engineering — Korea Advanced Institute of Science and Technology (KAIST), South Korea *Sep. 2016 – Jul. 2018*
Aerospace composites & structures · KAIST Full Merit Scholarship

B.Tech. in Aerospace Engineering — Amrita Vishwa Vidyapeetham, Coimbatore, India *Sep. 2011 – Jun. 2015*
3rd in departmental ranking · Published novel solvent bonding technique (J. Adhesion Sci. Tech., 2015)

SELECTED PUBLICATIONS (5 PEER-REVIEWED JOURNAL ARTICLES)

OR: Ankem, V.A., Desaulniers, G., Gamache, M., & Raymond, V. "New cutting planes for open-pit mine scheduling with multi-period block extraction." Computers & Operations Research, 2026. doi.org/10.1016/j.cor.2026.107408

OR: Ankem, V.A., et al. "A relax-and-repair framework and a parallel warm-start hybrid heuristic for open-pit mine production scheduling." Engineering Optimization (major revision), 2026.

AE: Kim, Y., Ankem, V.A., et al. Composite Structures, 2018 · Sathish Kumar, S.K., Ankem, V.A., et al. Composites Research, 2018 · Nam, Y.-W., Ankem, V.A., et al. Composite Structures, 2018 · Ankem, V.A., et al. J. Adhesion Sci. Tech., 2015.

Conferences: JOPT 2025, 2026 · ISMP 2024 · IISE 2024 · AACCM 2018 · KSCM 2017, 2018 · Advances in Composite Materials 2017 · ICTACEM 2014 (9 total)

GRANTS, FELLOWSHIPS & AWARDS

MITACS Fellowship — Ph.D. funding, Polytechnique Montréal & Rio Tinto · Sep. 2022

GIFAS Scholarship — ENAC, France · Sep. 2018

KAIST Full Merit Scholarship — graduate studies · Sep. 2016

Top 3 in B.Tech Aerospace Engineering Department — Amrita Vishwa Vidyapeetham · Mar. 2016

TECHNICAL SKILLS

Programming: C++, Python, R, SQL, SAS, C · **Solvers**: Gurobi, Xpress, Pyomo

Data & ML: PyTorch, Graph Neural Networks, NumPy, Pandas, scikit-learn, Tableau, Random Forest, Regression, Deep Reinforcement Learning (PPO, DQN, actor-critic, Q-learning), Gymnasium · **Tools**: Git, LaTeX, pytest

Languages: English (fluent), French (intermediate), Telugu (native), Hindi (fluent), Tamil (basic), Korean (basic).

LEADERSHIP & SERVICE

Student Representative & DEI Committee Member, GERAD (Sep. 2024 – Present) · Graduate Student Coordinator, ISSAC — KAIST (2017–18)

Director of Multimedia, KAIST International Students Association (2016–18) · ENAC Debate Club / MUN Lyon (2018–21)